Corporate Finance Assignment – Analysis of Nestle SA

February 12, 2016

Introduction/Background

Nestle is a multinational company offering products related to nutrition, health and wellness. Based in Switzerland, with a history of 150 years, Nestle had sales of SFr 91.6bn in 2014, selling over 2000 brands, operating in 197 countries, and with 339,000 employees.

I. Stockholder Analysis: Who has the power?

As of December 29 2015, the largest shareholder of Nestle stock is Blackrock, a global asset manager, with 3.7% of shares held (Source: Bloomberg, Dec 2015). The second-largest holder is surprisingly Nestle itself, holding a sizable 3.2% of the company's shares. According to the financial statement, these shares are used for the purpose of trading, as well as a share buy-back program and long-term incentive plans.
All top 20 stockholders (except Nestle) are institutional investors from the financial services industry in various regions: asset managers, universal banks and fund companies, making up a total of around 23% of total Nestle shares. Based on this, I assume the marginal investor in Nestle is well-diversified.

The board of directors is relatively large, with 14 members. There is also a considerable amount of insidership in the Nestle board of directors, with evidence that Nestle stockholders are not well represented: The current Chairman, Peter Brabeck, has been working for Nestle since 1968. In 1997, he became the CEO and a member of the board of directors at the same time! From 2005 to 2008, he acted as Chairman of the Board and CEO at the same time. In 2008, he stepped down as CEO, and has since then remained Chairman of the Board. Paul Bulcke, CEO since 2008, is also a member of the board of directors! For the residual 12 members of the board there is no direct evidence of insidership. Only 2–3 of them have a food and nutrition background, and 3–4 have a financial services background.
Given that the top 20 shareholders are all institutional investors, they may have very little interest in the management's actions and decisions. In addition, the Board of Directors is likely not challenging the management enough (on behalf of Nestle's shareholders), due to insidership described above.

To further improve its governance, I recommend that Nestle takes the following actions: (1) shrink the board of directors, (2) increase independence of the board by excluding the CEO and other insiders, (3) shrink the amount of own shares held, and (4) ensure that shareholders are better represented in the board, and that the board challenges Nestle's management frequently.

II. Risk and Return: How risky is Nestle and what is its hurdle rate?

[see last page for analysis sheet]

Comparing Nestle's stock performance to the Swiss market index (SMI) in 2013–2014 (Source: Bloomberg regression) reveals that Nestle shares outperformed the market by an annual return rate of **3.3% (Jensen's alpha).** The analysis also reveals a beta for Nestle that is very likely to be in the range of 0.72-0.93. 69% of the variation in Nestle's stock performance can be explained by the market variation (i.e. market risk), and 31% is firm-specific risk.
Based on this historical analysis and the Capital Asset Pricing Model (CAPM), one could expect Nestle's stock to yield a return of 6.18%.

Are you using weekly or monthly returns	W	
Current Riskfree Rate	-0.06%	
Risk Premium	7.57%	
Beta	0.824	
Std Error of Beta	0.054	
Intercept	0.063%	
Past Riskfree rate (Annual)	0.00%	
This is what your output shoud look like		
Rf (1- Beta)	0.00%	
Jensen's Alpha by period	0.063%	
Jensen's Alpha Annualized	3.33%	(Annual)
67% range	0.88	0.77
95% range	0.93	0.72
Expected Return	6.18%	

Figure: Results of historical stock performance analysis

Nestle is a multinational company producing mainly packaged food. Nestle reports its businesses by zone/region, and by the business segments Nestle Water, Nestle Nutrition (infant food) and "Other businesses" like Nespresso, Nestle Health Science and Nestle Skin Health. Based on this, I use industry averages from the food processing and the healthcare product industries to assess the overall risk of Nestle's business. Note that the healthcare product business is slightly riskier compared to food processing. Estimating 72% food processing and 28% healthcare product revenues, I come up with a bottom-up weighted-average **unlevered beta of 0.78** and a **levered beta of 0.84** for Nestle (its market debt-equity ratio is estimated at 10.0%, lease commitments included). Thus, Nestle operates with a risk profile below market risk.

Using a risk-free rate of –0.06% for the Swiss franc, and an estimated equity risk premium of 7.57% (based on regional ERPs weighted by regional revenues of Nestle), and a levered beta of 0.84, Nestle's **cost of equity is 6.31**%.

Nestle's pre-tax cost of debt is 0.94% (estimated based on Nestle's Aa2 rating and a corresponding default spread of 1%), and the marginal tax rate for Switzerland is 17.9%, yielding an **after-tax cost of debt for Nestle of 0.77%**.

With a market capitalization of around SFr 240 bn and a current market value of debt of around SFr 24 bn (including lease commitments), Nestle has a market debt-to-capital ratio of 9.1%, which yields a total **cost of capital of 5.8%**.

Nestle should only undertake investments that return at least 5.8% (the minimum acceptable hurdle rate).

Cost of capital	
Equity component	*Nestle*
Market value of equity	239608.3
Levered Beta	0.84
Riskfree Rate	-0.06%
Equity Risk Premium	7.57%
Cost of equity	6.31%
Debt component	
Market value of debt	$24'018.18
Pre-tax cost of debt	0.94%
Marginal tax rate	17.92%
After-tax cost of debt	0.77%
Market Debt to capital ratio	9.11%
Cost of capital	5.80%

Figure: Cost of capital calculation for Nestle

III. Investments: How good are Nestle's existing investments?

Nestle's investments likely vary across businesses and geographies, and I could not find a breakdown of Nestle's investment returns by business. However, as a firm, Nestle has earned an estimated **return on invested capital of 11.65%** in 2014 (estimate based on SFr 11.4 bn operating income and SFr 80.5 bn invested capital, adjusted for leases).

Comparing this with the current cost of capital of 5.8% yields a return spread of 5.85%, i.e. Nestle's recent/current investments yield a return that is almost 6% higher than Nestle's cost of capital. In my view, this is a very good performance upon which Nestle can build for the future.

In its annual report 2014, Nestle announced new investments and research collaborations to enhance its ability to deliver science-based solutions, e.g. a new research partnership with the EpiGen consortium to better understand the influence of nutrition and genetics at the beginning of life. Another example is the Nestlé Institute of Health Sciences' collaboration on a new project that "could one day lead to the development of made-to-measure vitamin combinations tailored to individual needs." Nestle says it "continues to invest in science-based innovation" to ensure that it can answer current and future needs.

Nestle appears to be investing further in various fields, with health science and genetics seeming to play a somehow larger role. While, with its current businesses, I think Nestle has a strong basis for further investment in the future, I would advise to carefully assess

the risks of these projects individually, since health-science-related projects are typically riskier compared to food-processing projects.

IV. Capital Structure Choices: What are Nestle's financing choices?

Nestle is currently financed with 90.9% equity (approx. SFr 240 bn) and 9.1% debt (approx. SFr 24 bn). It has short-term debt as well as long-term debt (with maturities beyond six years) in several major currencies. The average maturity of Nestle's debt is an estimated 3.55 years.

A certain level of debt for Nestle has the following advantages and disadvantages:

- Debt financing has a tax benefit since interest payments are tax-deductible. However, with a marginal tax rate of 17.9% for Switzerland, this advantage may be small compared to firms domiciled in other countries.

- Added discipline: A higher debt-to-capital ratio for Nestle may increase management's discipline and help the company to avoid bad investments and keep an eye on liquidity. Given its current governance setup (see Section I), it may be beneficial for increasing management accountability.

- Expected bankruptcy cost: Higher levels of debt may increase the probability of bankruptcy. However, Nestle has relatively stable earnings over time due to the nature of the food-processing business, and thus could afford higher levels of debt financing (compared to firms with more volatile earnings). Also, I would assess expected bankruptcy costs for Nestle to be lower compared to other firms due to the short lifetime of its products, i.e. higher levels of debt make sense for Nestle.

- Agency costs: I see Nestle as a firm where lenders can monitor its performance relatively easily due to its public/known products, which are present in almost everybody's life, i.e. Nestle should be able to borrow more compared to other firms.

- Loss of flexibility: Nestle is unlikely to experience a loss of flexibility when borrowing more. I would expect Nestle to be able to project its funding needs well, and thus be able to borrow more. Also, Nestle has excellent access to capital markets (given that the top 20 investors are all financial services firms).

All aspects described above indicate that Nestle would be in a good position to fund itself with a healthy level of debt.

V. Optimal Financing: What is the optimal mix of debt and equity for Nestle?

[see last page for analysis sheet]

Nestle's current debt-to-capital ratio is 9.1%, which is a relatively defensive leverage level for a food firm with relatively stable earnings. What would happen if Nestle were to change its debt-to-capital ratio, and which ratio would be optimum in terms of maximal firm value and minimal cost of capital?

Simulating across various debt-to-capital ratios, and taking to account changes in the levered beta, cost of equity, interest payments, interest coverage ratios, likely changes in bond ratings and cost of debt – the following is revealed:

By raising its debt-to-capital ratio up to around 30%, Nestle could increase its firm value from SFr 255 bn to SFr 260 bn, and its stock price from SFr 75.15 to SFr 76.65 (+ 2%). By doing this, Nestle's cost of capital would drop to 5.69%.

RESULTS FROM ANALYSIS			
	Current	*Optimal*	*Change*
D/(D+E) Ratio =	9.11%	30.00%	20.89%
Beta for the Stock =	0.84096059	1.05	0.21
Cost of Equity =	6.31%	7.89%	1.59%
Rating on Debt	Aa2		
After-tax cost of Debt =	0.77%	0.57%	-0.21%
WACC	5.80%	5.69%	-0.11%
Implied Growth Rate =	-0.06%		
Enterprise value	$254'745	$259'530	$4'784
Value/share (Perpetual Gr	$75.15	$76.65	$1.50

Figure: Result of optimal leverage analysis

To achieve this, Nestle should go from today's SFr 24 bn of debt (including lease commitments) to around SFr 79 bn, i.e. raise an additional SFr 55 bn.
This would obviously be a very bold move for Nestle. In any case, before implementing this massive change, I would advise Nestle
(a) to always ensure that all additional debt is tax-deductible, and
(b) to verify that, even in a bad year in terms of earnings, there would still be enough liquidity to cover (now higher) debt interest payments.

Due to the immense size of the additional debt capacity, Nestle should raise its debt level in gradual steps, and in doing so, keep a close look at its leverage ratios and the assessment of rating agencies (by any means avoid a downgrade by a ratings agency).

VI. Financing Changes: Should Nestle move to the optimal, and how?

How should Nestle raise additional debt of up to SFr 55 bn? In its current shape, with a market capitalization of SFr 240 bn and its recent over-performance, Nestle is hardly a takeover target.
And, the company's current investments yield a return on invested capital (ROIC) clearly above its current cost of capital (see Section III). Thus I recommend that Nestle takes on additional suitable projects with returns above its hurdle rate, and finances them with additional debt. This may include natural expansion or acquisitions in the same or related business areas.

I would advise Nestle to align the debt maturities with the durations of those projects (and the respective assets generating cash flows). I think projects in the food-processing business should be financed with shorter-term debt of similar maturities (probably 3–5 years) and currencies related to the expected cost and revenue streams.

For projects in the health-science area, I would assume longer lifecycles (5–10 years), and thus recommend financing them with longer-term debt, again with a mix of currencies related to the project costs and expected revenues.

The more uncertainty Nestle sees in the future and in its newly started projects, the more it should use floating-rate debt. And if there are projects that expect a high growth of cash flows generated, I think Nestle should finance them with convertible debt.

VII. Dividend Policy: How does Nestle return cash to stockholders?

[see last page for analysis sheet]

Nestle has returned cash to its shareholders every year (for the six years analyzed from 2009 to 2014), both in the form of dividends and share buybacks, at an average of SFr 10.5 bn per year.

Analysis of Past Dividends

	1	2	3	4	5	6	Aggregate
Net Income	$14'456.00	$10'015.00	$10'611.00	$9'487.00	$34'233.00	$10'428.00	**$89'230.00**
- (Cap. Exp - Depr)	($4'193.00)	$906.00	$13'125.00	$5'375.00	$3'432.00	$1'811.00	**$20'456.00**
- ∂ Working Capital	($114.00)	$1'360.00	$1'988.00	($1'983.00)	($632.00)	$2'442.00	**$3'061.00**
Free CF to Equity (pre-debt)	$18'763.00	$7'749.00	($4'502.00)	$6'095.00	$31'433.00	$6'175.00	**$65'713.00**
+ Net Debt Issued	($537.00)	($5'673.00)	$5'270.00	$2'207.00	($3'304.00)	$1'677.00	**($360.00)**
= Free CF to Equity (actual d	$18'226.00	$2'076.00	$768.00	$8'302.00	$28'129.00	$7'852.00	**$65'353.00**
Free CF to Equity (target debt	$18'370.60	$7'955.45	($3'125.10)	$6'404.03	$31'688.10	$6'562.48	**$67'855.56**

Dividends	$6'863.00	$6'552.00	$6'213.00	$5'939.00	$5'443.00	$5'047.00	**$36'057.00**
+ Equity Repurchases	$1'721.00	$481.00	$532.00	$5'480.00	$12'135.00	$7'013.00	**$27'362.00**
= Cash to Stockholders	$8'584.00	$7'033.00	$6'745.00	$11'419.00	$17'578.00	$12'060.00	**$63'419.00**
Dividend Ratios							
Payout Ratio	47.48%	65.42%	58.55%	62.60%	15.90%	48.40%	
Cash Paid as % of FCFE	47.10%	338.78%	878.26%	137.55%	62.49%	153.59%	

Figure: Free Cash Flow to Equity, Dividends, Equity Repurchases

Payout ratios (dividend payments/net income) were between 47% and 65% (with the exception of 2010 where Nestle had an outlying high net income). As a firm, Nestle is above average when it comes to payout ratios (compared to firms globally).

Over the past six years, Nestle's overall free cash flow to equity (FCFE) of SFr 65.4 bn matches closely with the total dividends paid to shareholders (SFr 63.4 bn), i.e. over the longer term, Nestle looks balanced as far as returning cash to shareholders. Looking at each year, interestingly, in four of the six years analyzed, Nestle paid more dividends than FCFE (i.e. compared to what it could have afforded in that particular year).

VIII. Dividend Assessment: Should Nestle return more or less cash to stockholders?

Nestle has been paying dividends and has bought back shares with quite a tradition over the past years. In 2014, it had over SFr 18 bn of FCFE, and paid SFr 8.6 bn back to shareholders. This corresponds to a considerable payout ratio of 47% (above average).

On the other hand, Nestle seems to have really good projects, with a return on invested capital (ROIC) of 11.6%, almost double its cost of capital. If Nestle is confident that its projects are as good as the ROIC shows, **Nestle should pay less dividends and invest additional cash flows in additional good projects**. When project returns become worse, Nestle should pay more dividends again.

For example, in 2014, if Nestle would have decided for a (slightly lower) 40% payout ratio (with which it would still be above average), it would still have paid SFr 5.8 bn in dividends/buybacks, but could have invested an additional SFr 2.8 bn into existing/new projects.

My other recommendation for Nestle would be to **avoid paying dividends above its FCFE** (which happened in four of the past six years). There are good reasons for Nestle to lower dividends if there is less FCFE, and this can also be communicated to shareholders with reasonable rationales. Shareholders should also notice if, for any reason, there is less cash flow in a particular year.

IX. Valuation: What is the value of Nestle and its equity?

[see last page for analysis sheet]

[Please note that this valuation was done on Feb 12 2016, i.e. before Nestle's publication of 2015 results]

The valuation of Nestle is based on many input parameters taken from the years 2014 and 2013, such as sales, EBIT, interest expense, book value of equity and debt, as well as cash.
The methodology is to estimate free cash flows to the firm (FCFF) over the next 10 years, including a terminal value.

The main drivers for the valuation model are (a) annual sales growth, (b) target EBIT, and (c) sales to capital ratio

For annual sales growth, I estimated a weighted global industry average of **8.21%**, assuming that Nestle has 75% of its sales in food processing and the rest in healthcare products. Note that Nestle's historical CAGR is -3.1%. However I chose to go with global industry averages, since Nestle's historical figure includes impacts of acquisitions and disposals of businesses (e.g. disposal of Alcon in 2010).

For the target EBIT margin, I chose the average over the past 6 years, **14.2%.** This reflects well how profitable Nestle has been in its operations.

For the sales-to-capital ratio, I again estimated a weighted average from the global industries in food processing and healthcare products, which is **1.58** (0.75*1.75 + 0.25*1.06)**.**

Using those parameters, the valuation model provides a **value for Nestle of SFr 240.8 bn**, and a **target share price of SFr 75.54**. Compared with the current price of 71.25 on Feb 12 2016, Nestle seems to be **undervalued by around 6%.**

Value of equity	$ 240'837.18
- Value of options	$0.00
Value of equity in common stock	$ 240'837.18
Number of shares	3'188.40
Estimated value /share	$ 75.54
Price	$ 71.25

(Of course, in a real situation I would wait for Nestle to release its newest figures, and then re-run the valuation with most updated input parameters.)

X. Overall Summary

1. Nestle is a very strong and established global food player, and a sound investment for shareholders

2. Nestle should review its governance, particularly with regard to insidership

3. Nestle seems to have a good portfolio of investments, earning higher returns compared to its current cost of capital

4. Nestle is currently not much leveraged, and could benefit from raising its leverage by investing in more good projects and funding them with additional debt

5. Nestle has been paying relatively high dividends. Nestle should continue to pay dividends, but not more than what it can afford in terms of FCFE. In years with excess FCFE, Nestle should invest excess cash in good projects

6. Nestle has an estimated value of SFr 240.8 bn, and an estimated target share price of SFr 75.54